SHEBIN T R

Backend, Systems & AI/Automation Engineer

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PROFESSIONAL SUMMARY

Backend and Systems Engineer with practical experience architecting high-concurrency microservices, distributed AI agent gateways, and cloud API platforms (Python, Go, Node.js, Redis). Proven track record engineering preemptive rate-limiting engines, document RAG ingestion pipelines, and transactional database schemas across Docker, Linux VPS, and cloud environments. Focused on building robust, idempotent, and defensible systems for backend, systems, and AI/automation roles.

TECHNICAL SKILLS

Languages & Core: Python, Go (Golang), JavaScript, TypeScript, React, SQL (PostgreSQL, SQLite), Object-Oriented Programming (OOP)
Backend & Microservices: FastAPI, Flask, Node.js, Express.js, AsyncIO, RESTful APIs, JSON-RPC, ONNX Runtime
Databases & Caching: Redis (Rate Limiting, Distributed Locking, Key-Value Caching), MongoDB, PostgreSQL, SQLite (WAL mode), VectorDB
AI & Automation: Google Vertex AI, Groq API, Claude, Local Models, Telethon (MTProto), Telegram Bot API, Retrieval-Augmented Generation (RAG), Vector Search
DevOps & Cloud: AWS, GCP, Linux VPS Administration (Ubuntu/Debian), Docker, Nginx (Reverse Proxy, DNS Routing, TLS), Git/GitHub CI/CD, Render, Vercel

FEATURED ENGINEERING PROJECTS

OrchestraAI Backend – Cloud-Native AI Agent BaaS Platform *Python 3.13+, FastAPI, AsyncIO, Redis, Docker, Nginx*

Core Architect & Open-Source Contributor

- Collaborated in designing an asynchronous BaaS runtime in FastAPI, abstracting hierarchical user/session memory and multi-provider model routing behind a unified API.
- Engineered a preemptive Zero-429 rate-limiting and key rotation engine using Redis atomic metrics, monitoring provider quotas to prevent throttling drops across multi-model key pools.

Non-Custodial Multi-Chain Crypto Payment Verification API *Node.js, Express, MongoDB, JSON-RPC, Render*

Backend Developer & Architect

- Developed a standalone non-custodial payment verification API on Render, interfacing with EVM/BSC (BNB) network via raw Alchemy -RPC node endpoints.
- Engineered asynchronous workers to poll on-chain transaction receipts, verify block confirmation depths, and execute idempotent transaction settlements in MongoDB.

Distributed Multi-Service Cloud Utility API Platform *Google auth, FastAPI, Flask, Node.js, Redis, MongoDB, ONNX, Vercel/Render*

Full-Stack Backend Engineer

- Architected a distributed utility platform across 5 cloud nodes for several APIs including cryptocurrency payment gateway with a central gateway managing token authentication, request logging, and Redis-backed IP rate limiting.
- Engineered a group-chat automated account anomaly detection API using feature extraction from chat data and deployed an ONNX-optimized model for lightweight, real-time inference.

Document RAG & Knowledge Ingestion Framework *Python, ChromaDB, Groq API, ML Document Classifier, JSON*

Backend & AI Engineer

- Engineered an automated ingestion pipeline converting unstructured PDF/DOCX files into semantically indexed chunks using ML heading/paragraph classification and ChromaDB.
- Integrated Groq LLM API with dynamic context summarization and token optimization, maintaining prompt payloads under 6k tokens for coherent multi-turn chat.

Autonomous Telegram Vertex AI Agent & Automation Engine *Python, Telethon, Telegram Bot API, Google Vertex AI, SQLite WAL*

Systems & Automation Developer

- Built an autonomous Telegram userbot using Telethon and Telegram Bot API to automate client interactions, environment inspection, and task scheduling during offline periods.
- Integrated Google Vertex AI for dynamic intent parsing and configured SQLite in WAL mode for persistent, lock-free session state storage during burst traffic.

WORK EXPERIENCE

Freelance Backend & Infrastructure Engineer | Independent Contractor *Remote*

- Delivered custom backend services, automation bots, and RESTful APIs for international clients using Python, Node.js, and React.
- Designed transactional database schemas in MongoDB, PostgreSQL, and SQLite, optimizing query performance and relational integrity.
- Benchmarked and refactored bottlenecked Python services to Go and asynchronous Node.js, reducing server memory footprints on Linux VPS nodes.
- Provisioned and maintained client cloud hosting and VPS deployments on Render, Vercel, and AWS, configuring Docker containers, reverse proxies, and CI/CD pipelines.

EDUCATION

Holy Grace Academy of Engineering | B.Tech in Artificial Intelligence & Data Science Engineering *Mala, Kerala, India | 2023 – 2027*

Core Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Artificial Intelligence (AI), Machine Learning & Deep Learning (ML/DL), Data Science.